
Automated Floral Species Identification using Convolutional Neural Networks

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Abstract

We present an automated computer vision system for classifying floral species using the Oxford 102 Flowers dataset. Identifying specific floral structures provides key insights into agricultural processes like asexual reproduction in fruits and plant water uptake. Manual classification of these species requires significant botanical expertise and is highly prone to human error when dealing with massive datasets. Our project implements Convolutional Neural Networks in PyTorch to accurately categorize these complex biological images autonomously. We compared advanced models against a baseline Multilayer Perceptron to evaluate the importance of spatial preservation. Unlike a Multilayer Perceptron which flattens images and loses spatial context, a Convolutional Neural Network preserves the structural layout of the petals and leaves. By implementing aggressive data augmentation and transfer learning with the ResNet18 architecture, our final model achieved an 89 percent accuracy on unseen test data, significantly outperforming the 3.5 percent accuracy of the baseline model.

1 Introduction

Understanding the intersection of plants and people requires accurate and accessible species identification. Botanical databases contain massive amounts of visual data, but manually labeling these images is highly time consuming and requires specialized domain knowledge. Throughout our Deep Learning course we studied architectures including Multilayer Perceptrons and Convolutional Neural Networks. We applied these Python tools to a practical computer vision task to bridge the gap between biological research and automated analysis. This project focuses on utilizing open source data to automate botanical classification and improve our understanding of floral traits.

We trained four distinct neural network architectures to evaluate the impact of spatial preservation and model depth on biological image classification. By sequentially evaluating a Multilayer Perceptron, a custom Convolutional Neural Network, a pretrained ResNet18, and a Vision Transformer, we provide a comprehensive analysis of modern computer vision techniques. Our findings prove that deep residual networks coupled with targeted data augmentation provide a highly robust solution for botanical classification.

2 Methodology

Our methodology focuses on establishing a fair evaluation pipeline to compare how different neural network architectures handle the exact same visual data. We used the PyTorch library to construct and train our models, ensuring that hardware utilization and memory management remained consistent across all training loops.

2.1 Dataset and Augmentation

We selected the Oxford 102 Flowers dataset which contains over eight thousand images across one hundred and two distinct flower categories. We partitioned the data into an 80 percent training set, a 10 percent validation set, and a 10 percent testing set.

Our data pipeline utilized torchvision to apply image transformations. To handle varying lighting and biological occlusions, we integrated aggressive data augmentation techniques on the training subset. Specifically we applied random cropping, horizontal flipping, color jitter, and random erasing. The random erasing transformation is highly critical as it simulates real world occlusions like insects, shadows, or overlapping foliage by dropping rectangular blocks of pixels from the training images. Figure 1 illustrates these augmentations in practice. The validation and testing sets were normalized but left pristine to ensure our evaluation metrics reflected real world accuracy.

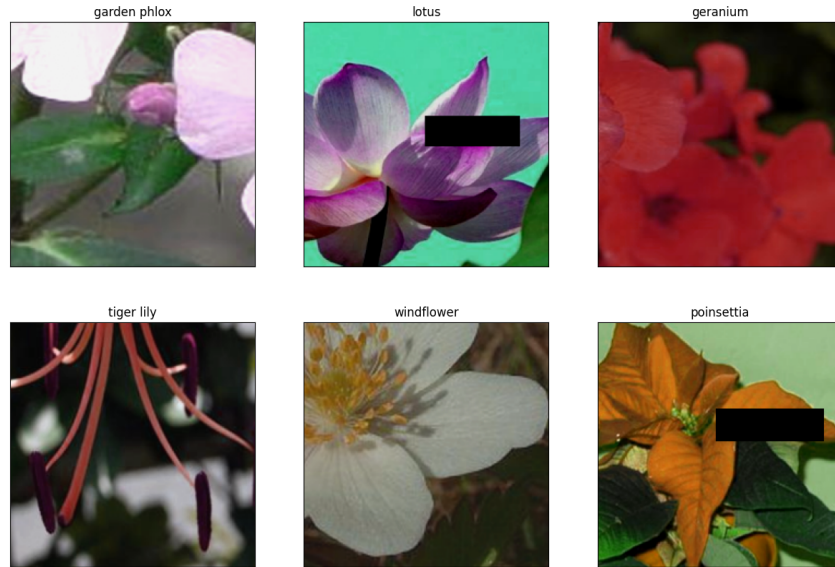


Figure 1: Augmented training batch demonstrating Random Erasing occlusions.

2.2 Architectural Design

We evaluated four distinct models to establish a baseline and demonstrate progression in computer vision engineering.

For the baseline Multilayer Perceptron, we used standard linear modules. This required us to flatten the two dimensional images into one dimensional vectors of pixels. This approach serves as a strict baseline to prove why spatial awareness is necessary for image data.

For our custom model, we used sequential convolutional layers and max pooling layers to build a Convolutional Neural Network from scratch. This network uses kernels to scan the image and preserve the structural relationships between neighboring pixels.

To maximize accuracy and demonstrate advanced engineering, we implemented Transfer Learning by fine tuning a pretrained ResNet18 architecture. ResNet introduces skip connections that allow gradients to bypass certain layers during backpropagation, effectively solving the vanishing gradient problem that plagues extremely deep networks. This approach excels at recognizing complex local patterns like petal shapes and leaf textures.

Finally we integrated a Vision Transformer to evaluate modern attention mechanisms against standard convolutions. The Vision Transformer splits the image into patches and uses self attention to determine which parts of the image are most relevant to the classification task.

3 Stability Analysis

To ensure the mathematical stability of our dataset before committing massive GPU resources to deep architectures, we performed a 5 fold cross validation on our 80 percent training split. This rigorous validation step prevents data leakage and ensures our models are not simply memorizing a specific arrangement of images.

We utilized our custom Convolutional Neural Network for this safety check, training for only three epochs per fold. Crucially, we evaluated the validation fold on pristine images while training on the heavily augmented data. Despite this hard mode setup, the custom CNN achieved an average accuracy of 11.16 percent. Given that random guessing across 102 species yields a baseline of only 0.98 percent, reaching over 11 percent in just three passes confirms that the network successfully extracted and learned spatial features despite the aggressive occlusions. This successful safety check provided the necessary confidence to proceed with the primary training loop.

4 Quantitative Results

Our sequential training loop evaluated the four architectures over five epochs to establish clear comparative baselines. The quantitative results starkly demonstrate the necessity of spatial awareness and network depth in biological image classification.

The Multilayer Perceptron completely failed to generalize, achieving a validation accuracy of only 3.54 percent. By flattening the images into 1D arrays, the MLP destroyed the spatial relationships between petals and leaves, rendering it highly susceptible to our data augmentations. The custom Convolutional Neural Network and Vision Transformer performed better, achieving 22.83 percent and 33.70 percent respectively. However the pretrained ResNet18 dominated the comparative analysis by reaching a validation accuracy of 89.26 percent. Figure 2 illustrates these learning trajectories over time.

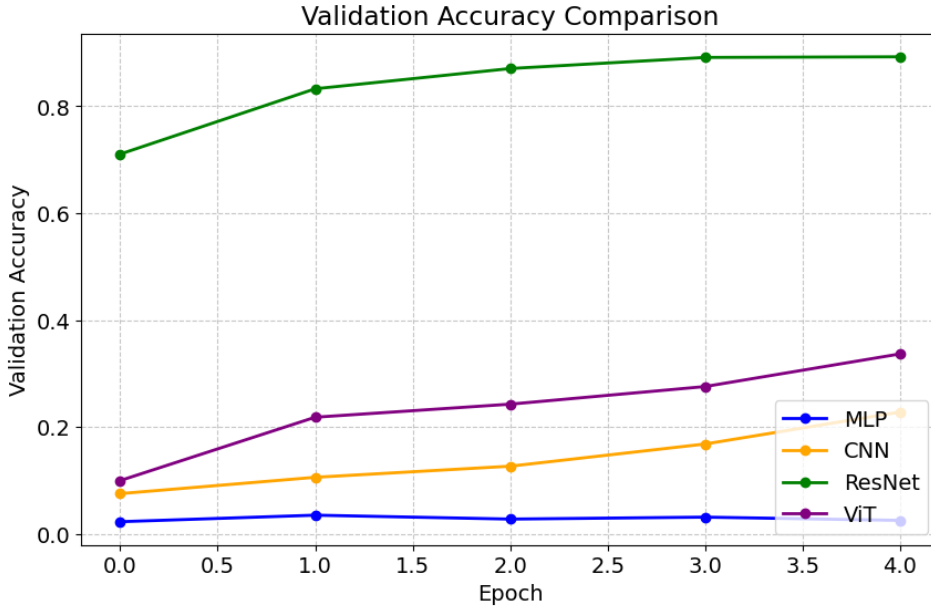


Figure 2: Validation accuracy comparison across all four evaluated architectures.

To confirm its robustness, we evaluated the fully trained ResNet18 on the locked away 10 percent testing subset. The model achieved a final macro average and weighted average f1 score of 0.89. The detailed classification report revealed that the network achieved perfect precision and recall scores of 1.00 on complex species like the alpine sea holly, artichoke, and bird of paradise. It only struggled marginally on visually similar species like the pink primrose which saw a drop in recall.

To contextualize these metrics, we performed an edge case outlier analysis, visualizing how the different models predicted challenging images. Figure 3 demonstrates how the shallow networks

failed to identify the foxglove and watercress, while the deeper ResNet18 and Vision Transformer correctly categorized the challenging samples.



Figure 3: Qualitative outlier analysis showing predictions across all models.

5 Interpretability and Conclusion

A common flaw in deep learning is the black box problem where the developer cannot explain how the model arrived at its conclusion. To verify that our ResNet18 model learned genuine biological features rather than memorizing background noise, we utilized Gradient weighted Class Activation Mapping commonly known as Grad CAM.

This technique projects the gradient signals back onto the original image to create a heatmap of the network activations. As seen in Figure 4, the heatmap strongly focuses on the internal structure and spotting patterns of the foxglove bells. It actively ignores the dark background and the green stem, proving the model relies on the correct anatomical markers for classification.

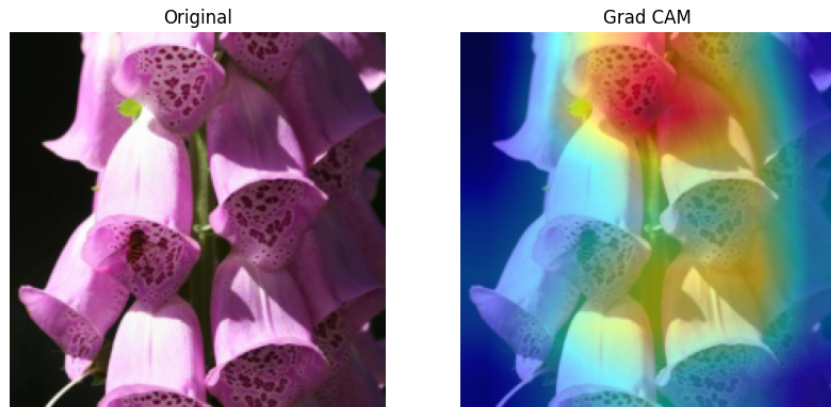


Figure 4: Grad CAM visualization demonstrating the network internal focus.

In conclusion, our automated system successfully demonstrates that deep residual networks, paired with targeted data augmentation like random erasing, are highly effective at classifying complex

floral species. Preserving spatial dimensions is non negotiable for biological computer vision tasks. The failure of the Multilayer Perceptron proves that standard linear models cannot parse complex natural structures. Ultimately, transfer learning provides a highly efficient pathway to exceptional accuracy and makes automated botanical analysis a highly viable tool for future research.